

Thermal Change, Network Fragmentation and Biotic Structure in a Mediterranean Fish Metacommunity

Final integrated report for the Guadalquivir River Basin
2026–2045 simulation horizon

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1 Abstract

Mediterranean river fish communities experience warming, dendritic habitat fragmentation and biological invasion simultaneously. Their responses cannot be inferred from climate forcing alone because persistence and thermal tracking depend on network connectivity and biotic interactions. We analysed a deterministic, spatially explicit metacommunity model of the Guadalquivir River Basin, southern Spain. The model represented 775 sites, 24 fish species, daily site-level water temperature, species-specific thermal performance, asymmetric interactions and directional dispersal through a dendritic network. We synthesised a climate ensemble of 44 combinations of 11 general circulation models and four Shared Socioeconomic Pathways, a no-warming control, a 32-run obstacle factorial experiment and a 48-run interaction-structure experiment.

Basin-mean warming by 2045 was 0.74–0.97 °C across emission scenarios, with individual climate-model combinations spanning approximately 0.28–1.37 °C. Total biomass declined weakly with warming (slope -0.120 model units per °C, $R^2 = 0.25$), while native richness was effectively unchanged (slope -0.00071 species per site per °C, $R^2 = 0.02$). The weak aggregate response resulted from opposing species-level thermal responses: dominant native cyprinids had optima above current basin temperature, whereas brown trout (*Salmo trutta*) was already on the warm side of its optimum. Trout biomass declined by 9.0–10.9% across scenarios, compared with 0.2% in the control, and declined at approximately 9.2% of baseline biomass per °C of warming ($R^2 = 0.65$).

In the obstacle experiment, upstream movement cost was the principal control of aggregate native biomass, whereas the climate endpoint was the principal control of trout biomass. Passability had a non-monotonic effect: blocking barriers reduced native biomass at low background upstream cost but increased basin-mean native biomass at high upstream cost. An invasive-favouring interaction matrix reduced native biomass by approximately 90% and increased invasive biomass nearly tenfold. This was a deliberately extreme, confounded contrast rather than an empirical uncertainty estimate.

The results indicate that, over this early-warming horizon, community-level indicators are more strongly controlled by network hydraulics and interaction structure than by approximately 1 °C of warming. Warming nevertheless produces an early decline in a cold-water specialist. Connectivity interventions should therefore be evaluated jointly with background network resistance, species composition and invasive pressure rather than under the assumption that increased passability always improves aggregate native outcomes.

2 Introduction

2.1 Metacommunities in dendritic river networks

Riverine metacommunities are structured by the interaction of local environmental filtering, demographic regulation, biotic interactions and dispersal through branching networks. The metacommunity framework motivates this synthesis [Leibold et al., 2004].

Dendritic topology constrains movement to connected channels and makes directional movement important [Altermatt, 2013]. Barriers can alter both the amount of movement and the spatial distribution of demographic rescue and colonisation [Perkin and Gido, 2012]. The effect of a barrier is therefore expected to depend on its position in the network, the background cost of upstream movement and the species using the corridor.

Mediterranean rivers are especially informative systems for examining these interactions. They contain range-restricted endemic fishes and strong spatial gradients in temperature, habitat quality and flow regulation. Warming can move species away from or towards their thermal optima, while fragmentation can prevent populations from tracking suitable reaches. Invasive species can change the direction and magnitude of these responses through asymmetric competition and predation.

2.2 The Guadalquivir Basin as a model system

The Guadalquivir River Basin is a large, strongly regulated Mediterranean basin in southern Spain with high Iberian freshwater endemism and a long history of dam-associated fish-community change [Herrera et al., 2022, Ramos-Merchante and Prenda, 2018]. The model domain contains 775 sampling sites aggregated into 774 water bodies and connected as a dendritic river network. The modelled assemblage contains 10 native residents, 10 invasive species and four migratory or diadromous taxa. Migratory taxa remain in the dynamical system and contribute to total biomass, but are excluded from native-richness reporting because their estuarine or marine life histories are not represented by the modelled freshwater corridor.

The basin contains a dense inventory of transverse structures. Of 1,658 non-completely-passable obstacles in the 2026 inventory, 973 were spatially matched to modelled network segments within 2,000 m. Matched links have base upstream passability of 0.1 and downstream passability of 0.5. A global upstream movement cost additionally reduces dispersal as elevation gain increases. These features make the basin a useful system for examining thermal-hydraulic coupling in a regulated Mediterranean river network [Perkin and Gido, 2012, Larsen et al., 2021].

2.3 Objectives

This report addresses three questions:

1. How strongly does projected warming between 2026 and 2045 alter aggregate biomass and richness relative to a stationary seasonal control?
2. How do upstream movement cost and local obstacle passability influence native, invasive and total biomass across the dendritic network?
3. Does the structure of biotic interactions alter apparent sensitivity to climate and connectivity, and which species provides the earliest indication of thermal change?

The analysis uses three complementary experiments: a multi-model climate ensemble, a balanced obstacle factorial and a deliberately contrasting interaction-structure sweep. The distinction between robust mechanistic results and effects conditional on perturbed model structure is maintained throughout.

3 Materials and Methods

3.1 Study domain and assemblage

The model represents one Guadalquivir basin with 775 sites, 774 reporting water bodies and 24 fish species. Native residents are *Aphanius baeticus*, *Anaocypris hispanica*, *Squalius pyrenaicus*, *Pseudochondrostoma willkommii*, *Luciobarbus sclateri*, *Squalius alburnoides*, *Iberochondrostoma lemmingii*, *Cobitis paludica*, *Iberochondrostoma oretanum* and *Salmo trutta*. Invasive species are *Gambusia holbrooki*, *Micropterus salmoides*, *Lepomis gibbosus*, *Cyprinus carpio*, *Carassius gibelio*, *Ameiurus melas*, *Oncorhynchus mykiss*, *Esox lucius*, *Gobio lozanoi* and *Tinca tinca*. Migratory or diadromous taxa are *Anguilla anguilla*, *Alburnus alburnus*, *Liza ramada* and *Mugil cephalus*.

The classification of *Alburnus alburnus* requires verification. It is grouped as migratory in the current model although source inventory information identifies it as exotic. This issue affects interpretation of invasive richness and biomass but does not affect the trout result or the hydraulic rankings.

3.2 Local population dynamics

For site i and species s , density $N_{i,s}$ follows

$$\frac{dN_{i,s}}{dt} = N_{i,s} \left[r_s W_{i,s} \left(1 - \frac{\sum_j N_{i,j}}{K_i} \right) + \frac{\sum_j \alpha_{sj} N_{i,j}}{K_i} - m_{i,s}^{\text{heat}}(t) \right] + \sum_{j \in \text{nb}(i)} (m_{ji}^s N_{j,s} - m_{ij}^s N_{i,s}). \quad (1)$$

The local term combines density dependence, environmental suitability and an asymmetric interaction matrix. K_i is site carrying capacity, r_s is the intrinsic growth rate and α_{sj} is the effect of species j on species s . Carrying capacity was scaled to 1 times observed total density with a minimum-capacity floor; the mean modelled capacity was approximately 86.6 compared with an observed mean of approximately 84.4. Species observed at a site received their full growth rate and absent species received 10% of the rate to permit colonisation. Annual literature-derived rates were converted to daily rates.

Environmental suitability was defined as

$$W_{i,s}(t) = \exp \left[-\frac{(T_i(t) - T_s^*)^2}{2\sigma_s^2} \right] h_i, \quad (2)$$

where $T_i(t)$ is daily site temperature, T_s^* is the species thermal optimum, σ_s is thermal breadth and h_i is a static habitat-suitability index bounded between 0.1 and 1.0. The habitat index was derived from the trophic-state index and held constant. Thermal optima were set to the midpoint of each empirical temperature range and thermal breadth to one-sixth of that range. The model therefore uses a fixed, symmetric Gaussian niche with no acclimation, plasticity or evolution.

Heat stress was represented separately:

$$m_{i,s}^{\text{heat}}(t) = k \max(0, T_i(t) - T_s^{\text{up}})^2, \quad (3)$$

where T_s^{up} is the empirical upper thermal limit. The coefficient is $k = 3.7817692640400324 \times 10^{-5}$ with units $\text{day}^{-1} \text{ } ^\circ\text{C}^{-2}$. It was calibrated so that the most exposed species–site combination under baseline climate lost no more than 5% of annual survival. No cold-stress mortality term was applied.

3.3 Dendritic dispersal and obstacles

Movement from site i to site j was defined as

$$m_{ij} = D_{\text{median}} \frac{x_{ij} p_{ij}}{d_{ij}}, \quad (4)$$

where $D_{\text{median}} \approx 0.0274 \text{ km day}^{-1}$, d_{ij} is hydrological distance, x_{ij} is the elevation-dependent movement factor and p_{ij} is directional passability. Species-specific rates were obtained by multiplying this base rate by literature-derived relative dispersal scalars. Migratory taxa generally had values above five times the median and sedentary taxa below 0.1 times the median.

Downstream or level movement was assigned $x_{ij} = 1$. For upstream movement,

$$x_{ij} = \frac{1}{1 + c\Delta e}, \quad (5)$$

where c is global upstream movement cost and Δe is elevation gain. The network was constructed from hydrological distances by connecting adjacent sites within sub-catchments and linking outlet sites across sub-catchments. The effective passability was the element-wise minimum of legacy dam and directional obstacle constraints. A sparse dispersal matrix was recomputed

for each upstream-cost and passability combination. Equal-elevation links were unrestricted by the obstacle overlay because flow direction could not be inferred.

The four passability states were baseline (multiplier 1.0), improved (1.5), reduced (0.5) and blocked (0.1), with passability capped at 1.0. No exploitation-pressure scenario was applied in the experiments reported here. Invasive responses are reported only as aggregate invasive biomass and richness; species-level invasive responses are outside the scope of this synthesis.

3.4 Temperature forcing and initial conditions

Climate simulations used daily site-level absolute water temperature for 11 general circulation models under SSP1-2.6, SSP2-4.5, SSP3-7.0 and SSP5-8.5. Anomalies relative to the 1986–2005 baseline were added to observed site temperatures over 2026–2045. The no-warming control repeated the baseline seasonal climatology. Elevation scaling of the warming anomaly was disabled, so all sites received the same basin-scale anomaly while retaining their different absolute temperature levels.

The climate ensemble and obstacle experiment used a converged 373-year seasonal baseline burn-in, with basin convergence tolerance 5×10^{-5} . The alternative-interaction experiment used a separate 21-year burn-in for each interaction matrix. The system was integrated with the adaptive fifth-order Runge–Kutta method ‘Tsit5’ using relative and absolute tolerances of 10^{-6} , with one time unit equal to one day and a positivity callback. Climate outputs were saved on a monthly grid and reported annually.

Two warming quantities were kept distinct. Climate dose–response analysis used imposed basin-mean warming at 2045. The obstacle experiment selected the coolest endpoint (SSP1-2.6/IITM-ESM) and warmest endpoint (SSP2-4.5/UKESM1-0-LL) using the maximum across sites of the final-year annual-mean anomaly; those endpoint values were approximately 0.856 °C and 1.852 °C, respectively. These quantities are not interchangeable.

3.5 Experimental design

Table 1: Current simulation design.

Component	Specification
Spatial domain	Guadalquivir basin; 775 sites and 774 water bodies
Assemblage	24 species: 10 native residents, 10 invasive and 4 migratory/diadromous
Climate ensemble	11 climate models $\times$ 4 SSPs = 44 scenarios plus one no-warming control
Obstacle experiment	2 climate endpoints $\times$ 4 upstream costs $\times$ 4 passability states = 32 runs
Interaction experiment	3 interaction matrices $\times$ 2 climate endpoints $\times$ 2 upstream costs $\times$ 4 passability states = 48 runs
Upstream costs	0.01, 0.05, 0.10 and 0.50 in the obstacle experiment; 0.01 and 0.50 in the interaction experiment
Interaction matrices	Original qualitative matrix; random placebo with off-diagonal values in $[-1, 0]$; invasive-favouring worst-case matrix
Simulation horizon	2026–2045 with daily temperature forcing and annual reporting

The original interaction matrix mapped qualitative categories to coefficients of -1.0 (no coexistence), -0.8 (displacement), -0.5 (predation), -0.3 (competition or interference), -0.2 (weak effect) and 0.0 (coexistence or neutrality). The random placebo used a fixed seed of 42. The invasive-favouring matrix imposed invasive-to-native effects of -0.8 , no native-to-invasive effects,

invasive–invasive competition of -0.3 and native–native competition of -0.1 . All alternative-interaction runs also used a thermal-breadth multiplier of 0.3 , so interaction structure, thermal breadth and burn-in duration are confounded in that experiment.

3.6 Response metrics and analysis

Richness was the number of species exceeding 0.1 density units, averaged over sites or reporting units. Native, invasive and total biomass were sums of the relevant species densities. Realised native richness loss was calculated as

$$L(t) = \max \left(0, 1 - \frac{R(t)}{R_0} \right), \quad (6)$$

where R_0 is richness in the spun-up baseline. This is not an extinction probability. The quasi-extinction diagnostic flagged a site when a species remained below $\max(0.1, 0.1 \times \text{baseline density})$ for three consecutive annual observations. It was not used as a primary climate outcome because the trout fraction was already near saturation.

Climate dose–response relationships were estimated by ordinary least squares of final-year basin metrics against imposed basin-mean warming across the 44 scenario runs, excluding the control. Factor effect sizes were calculated as the median within-block span over factor levels, standardised by the run-set standard deviation. Balanced within-experiment sums of squares were used descriptively. No formal significance testing was performed and no stochastic replicate simulations were run.

4 Results

4.1 Climate forcing and aggregate community response

Added basin-mean warming by 2045 was 0.74°C under SSP1-2.6, 0.87°C under SSP2-4.5, 0.88°C under SSP3-7.0 and 0.97°C under SSP5-8.5. Across individual climate-model and pathway combinations, the basin-mean warming axis spanned approximately 0.28 – 1.37°C . Climate-model spread overlapped differences among pathways.

The control was nearly stationary over 20 years: total biomass changed by $+0.08\%$, native biomass by $+0.10\%$ and native richness by -0.57% . The small residual drift is interpreted as an internal cycle rather than a climate response. Across climate scenarios, total biomass declined by -0.120 model units per $^\circ\text{C}$ ($R^2 = 0.25$), an effect of approximately 0.13% of the roughly 92.7 model units per site at the warming extremes. Native richness declined by only -0.00071 species per site per $^\circ\text{C}$ ($R^2 = 0.02$). The richness-loss indicator had slope $+0.00015$ per $^\circ\text{C}$ ($R^2 = 0.01$) and never exceeded approximately 0.9% .

Most dominant native cyprinids had optima of 19 – 20°C against a basin mean of approximately 16.0°C . Warming therefore moved their suitability towards the optimum. Brown trout had an optimum of 12.0°C and was already approximately 1.5 thermal standard deviations on the warm side of its optimum. Aggregate metrics consequently concealed opposing species responses.

4.2 Brown trout as a thermal–hydraulic sentinel

Brown trout was the only species with a large, directionally reproducible climate response. Basin trout biomass declined by 9.0% under SSP1-2.6, 9.7% under SSP2-4.5, 9.5% under SSP3-7.0 and 10.9% under SSP5-8.5, compared with a 0.2% decline in the control. Individual climate-model responses spanned roughly 4 – 19% declines. Across the ensemble, trout biomass declined by approximately 9.2% of baseline biomass per $^\circ\text{C}$ of warming ($R^2 = 0.65$). At occupied sites, median population size fell to approximately 68 – 92% of its 2026 value.

Thermal exposure increased with warming. The maximum number of days per year above the 20 °C upper limit increased from 131 under baseline conditions to 139–174 across scenarios. Maximum squared exceedance energy increased from approximately 1.6×10^3 to $2.1\text{--}4.2 \times 10^3$ °C² day. Trout occupies only approximately 4% of sites, so its large relative decline is diluted in basin-wide richness and biomass. It is an early-warning indicator of thermal–hydraulic coupling, not a proxy for whole-community vulnerability.

In the obstacle experiment, trout remained climate-dominated but was also sensitive to network structure. At upstream cost 0.05, final trout biomass declined from 803.9 to 770.9 model units when passability changed from baseline to blocked under the cool endpoint, and from 725.0 to 704.2 units under the warm endpoint. At the warm endpoint, relative changes were −13.6% (baseline), −12.8% (improved), −14.7% (reduced) and −16.1% (blocked).

4.3 Hydraulic controls and the passability sign reversal

Across eight response metrics, mean factor ranking was interaction matrix (1.88) > upstream cost (3.00) $\approx$ thermal breadth (3.12) > passability (3.62) > passability $\times$ upstream cost (4.38) > climate endpoint (5.00) > passability $\times$ temperature (7.00). The interaction matrix also had the largest mean normalised effect (0.871). The passability $\times$ temperature interaction had a median standardised magnitude no greater than 0.15 standard deviations for every metric except the near-saturated trout quasi-extinction flag, where it reached 1.24.

For native biomass, upstream cost explained $SS = 0.578$, followed by passability $\times$ upstream cost ($SS = 0.242$), passability ($SS = 0.155$) and climate endpoint ($SS = 0.026$). Trout final biomass had a different ordering: climate endpoint $SS = 0.886$, passability $SS = 0.090$ and upstream cost $SS = 0.011$. Total biomass and native richness were dominated by upstream cost, with $SS = 0.961$ and 0.947 , respectively. Invasive biomass also responded primarily to upstream cost ($SS = 0.970$).

The passability effect changed sign with background upstream cost. At the cool endpoint, the blocked-minus-baseline difference in basin native biomass was −0.349, −0.009, +0.146 and +0.442 at upstream costs 0.01, 0.05, 0.10 and 0.50. At the warm endpoint the corresponding values ranged from −0.336 to +0.453. Reduced passability showed the same reversal, from approximately −0.101 to +0.371 across the cool-end cost range, whereas improved passability reduced basin native biomass at every upstream cost, from −0.034 at cost 0.01 to −0.407 at cost 0.50.

This reversal was spatially heterogeneous. The positive basin-level blocked-minus-baseline effect at high upstream cost was a net result of positive and negative changes among sub-catchments and water bodies. Blocking did not uniformly benefit native fish; it redistributed biomass across the network. The outputs cannot uniquely separate loss of native immigration, reduced emigration and suppression of invasive dispersal. The robust finding is that the sign of a passability intervention depends on background network resistance.

4.4 Interaction structure and native–invasive redistribution

The alternative-interaction experiment produced the largest structural contrast. The invasive-favouring matrix reduced pooled native biomass from approximately 50.2 to 5.1 model units, native richness from 2.42 to 0.15 species per site and trout biomass from approximately 548 to 313 units. In the matched cool-end, low-cost, baseline-passability contrast, native biomass changed from approximately 51.0 to 5.2 units and trout biomass from approximately 575 to 314 units. Invasive biomass increased from approximately 6.0 to 58.7 units, while total biomass changed comparatively little, from approximately 56.2 to 64.2 units.

The interaction matrix accounted for $SS = 0.998$ of native biomass and $SS = 0.982$ of trout biomass in the alternative sweep. The confounded thermal-breadth contrast shifted native biomass downward by approximately 36.9–38.8% and trout biomass by approximately 27.0–

30.3%, but these changes cannot be attributed to thermal breadth alone. Passability spans were interaction-dependent: trout spans were largest under the invasive-favouring matrix (mean span approximately 23.8 model units, versus 10.2 under the original matrix and 8.5 under the random placebo), while native-biomass spans were smallest under the invasive-favouring matrix (approximately 0.2 units, versus 1.3 and 1.4 under the original and random matrices) after native biomass had already collapsed. The primary obstacle-sweep sign reversal was not recovered consistently in the alternative sweep.

4.5 Other species and aggregate indicators

Warm-adapted native cyprinids and most invasive and migratory taxa changed little over the study horizon. Their empirical upper thermal limits were generally 25–30 °C and were rarely exceeded, so the heat-stress mortality term was inactive. No warming-driven invasive expansion was detected. Invasive richness and total richness were effectively invariant across scenarios and the control, with final-year basin means of approximately 0.217 and 2.72, respectively.

5 Discussion

5.1 Warming was weak at the aggregate scale but strong for a specialist

The climate ensemble sampled an early-warming interval. Approximately 1 °C of warming by 2045 was insufficient to produce a large change in aggregate richness because most of the native assemblage was cold-marginal relative to its assigned optimum. This does not imply thermal safety. It reflects cancellation between warm-adapted dominant natives and a cold-water specialist moving away from its optimum.

The trout response was supported by three convergent diagnostics: monotonic biomass decline across climate models, separation from the stationary control and increased exposure above the empirical upper thermal limit. Its magnitude remains conditional on the Gaussian niche, midpoint optimum and unvaried thermal breadth. No basin-scale extinction forecast should be inferred from this response.

5.2 Connectivity is context dependent

The passability sign reversal challenges a simple interpretation of connectivity restoration as uniformly beneficial to native biomass. At low upstream movement cost, blocking a link removed movement and reduced native biomass. At high cost, further blocking was associated with higher basin-average native biomass, plausibly because it restricted movement by competitively effective invasive taxa or reduced net losses from isolated native units. The present design cannot discriminate among these mechanisms.

Barrier decisions should therefore be evaluated at reach and sub-catchment scales against background hydraulic resistance, species composition and invasive pressure. Basin means conceal local gains and losses. The positive aggregate effect at high cost was a spatial redistribution rather than a uniform improvement.

5.3 Interaction structure is a major structural uncertainty

The interaction matrix determined whether biomass was retained by native or invasive compartments in the deliberately perturbed experiment. This focus on interactions is consistent with broader work showing that climate sensitivity and invasion outcomes can depend strongly on biotic context [Carosi et al., 2023]. The invasive-favouring matrix is a bounding scenario, not an empirical uncertainty distribution. The random matrix is a placebo and its relative biomass

ratios are uninterpretable because some species have near-zero burn-in baselines. Future inference should use empirically estimated interaction coefficients, uncertainty distributions and independent replication across interaction and thermal-parameter sets.

5.4 Monitoring and management implications

Monitoring based only on aggregate richness or biomass is unlikely to detect early thermal deterioration. Cold-water specialists should be monitored using species-level abundance, occupancy, upper-limit exceedance and thermal selection gradients. Brown trout is a useful sentinel because warming and restricted movement act in the same direction for a sparse, headwater-associated population.

Connectivity interventions should combine local passability, upstream movement cost, species movement traits and invasive distributions. Barrier removal may provide demographic rescue in some reaches but may also facilitate invasive expansion or redistribute biomass away from vulnerable native units. Thermal refugia, especially high-elevation and headwater reaches, should be protected alongside connectivity improvements, consistent with adaptive river-management recommendations for uncertain futures [Tonkin et al., 2018].

Before quantitative management forecasts are attempted, the model should be extended to represent discharge and drought dynamics, abstraction, flow-dependent habitat change, land-use degradation, age structure, harvesting, acclimation and evolution.

6 Limitations and Scope of Inference

1. The climate and sensitivity experiments are separate designs. Each sweep is internally balanced, but factors cannot be pooled into one orthogonal analysis.
2. The thermal-breadth contrast is confounded with interaction structure and burn-in duration and is not an isolated estimate of thermal-breadth sensitivity.
3. The random interaction matrix is a placebo; its relative biomass ratios are not ecologically interpretable.
4. The invasive-favouring matrix is deliberately extreme and should be treated as a bounding scenario rather than a forecast or probability distribution.
5. No formal significance tests or stochastic replicate simulations were performed. Biomass is expressed in model units and is not calibrated field biomass.
6. Thermal optima and breadths were derived heuristically from empirical ranges. The mid-point optimum was not varied, so response magnitudes and potentially the aggregate balance require an optimum-position sensitivity analysis.
7. The model assumes fixed, symmetric thermal niches and excludes evolution, acclimation and plasticity.
8. All sites received the same basin-scale warming anomaly; spatial differences in warming among elevation zones and reaches were not resolved.
9. Discharge, drought, abstraction, land-use change and flow-dependent habitat quality were not dynamically represented.
10. The richness-loss indicator is realised threshold-based loss, not an extinction probability. Trout quasi-extinction was near saturation (approximately 0.956–0.959 in the obstacle sweep and 0.935–0.972 in the alternative sweep) and was not a primary climate outcome. No warming-driven species extinction was demonstrated.

11. The 2026–2045 horizon cannot robustly rank emission pathways because climate-model spread overlaps pathway differences. Absolute temperature changes relative to 1986–2005 are not directly comparable across scenario products because of different pre-2026 offsets.
12. Of the 1,658 inventoried obstacles, 685 were not matched to a modelled segment and did not affect the simulated dispersal field.
13. The baseline retains a small internal cycle; control changes below approximately 1% over 20 years are not interpretable as climate effects.
14. The classification of *Alburnus alburnus* requires verification before invasive richness and biomass are used for policy inference.

7 Conclusions

For the Guadalquivir metacommunity and 2026–2045 horizon, aggregate native biomass and richness were controlled primarily by network hydraulics and interaction structure, not by the approximately 1 °C of warming delivered by the climate ensemble. This weak aggregate response was not evidence of thermal safety; it resulted from cancellation between warm-adapted dominant natives and a cold-water specialist moving away from its optimum.

Brown trout provided the clearest early-warning signal, with a reproducible 9–11% climate-associated biomass decline and increased exposure to supra-optimal temperatures. Upstream movement cost was the main control of aggregate native biomass and richness, while climate was the main control of trout biomass. Passability effects were non-monotonic and changed sign with background upstream cost.

The strongest structural contrast arose from the invasive-favouring interaction matrix, which shifted biomass from native to invasive compartments while approximately conserving total biomass. Because this contrast was deliberately extreme and confounded with other design changes, it identifies a priority for empirical uncertainty reduction rather than a direct forecast. A defensible management strategy combines species-level thermal monitoring, protection of cold-water refugia, spatially targeted connectivity assessment and invasive-species control.

8 Acknowledgements

Fish sampling and handling were conducted under the relevant Spanish regulatory framework. Funding, permits, institutional support and author-specific contributions should be completed before formal publication.

9 Data and Code Availability

The study is based on empirical fish-density and juvenile observations, site connectivity and environmental measurements, species trait data, qualitative interaction information, a 2026 obstacle inventory and daily downscaled water-temperature projections for the Guadalquivir basin. The final release should provide a persistent accession for model code, input data, scenario forcing, derived outputs and figure source data, together with the applicable licence and any restrictions on sensitive spatial data.

10 Figures

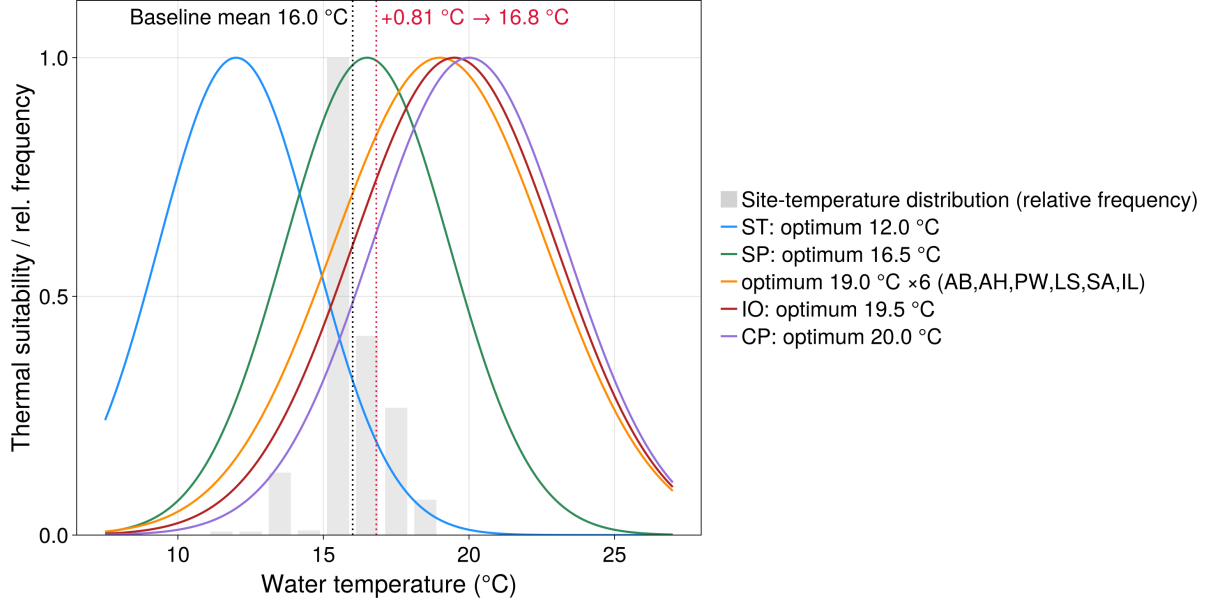


Figure 1: **Thermal niches against baseline water temperature (single panel).** *Grey bars:* relative frequency of the 2026 baseline water temperature across all 775 sampling sites; each bar is 1 °C wide and the counts are divided by the largest count, so the tallest bar equals 1. *Coloured curves:* the Gaussian thermal-suitability function $W = \exp[-(T - T_s^*)^2 / (2\sigma_s^2)]$ for the native species; species with identical empirical temperature ranges are drawn once and listed together. In the legend each colour gives a species code and its thermal optimum T_s^* ; because σ_s is one-sixth of the empirical range, suitability falls to $\exp(-9/2) \approx 0.011$ at the range edges. *Vertical lines:* dotted black is the basin-mean baseline temperature (16.0 °C); dotted crimson is the same mean after the median imposed warming to 2045 (+0.81 °C, i.e. 16.8 °C). *Reading:* the dominant cyprinids peak at 19–20 °C, warm of the present mean, so modest warming moves them towards their optima, whereas brown trout (*Salmo trutta*, ST, optimum 12.0 °C) is already about 1.5 thermal standard deviations (about 1.9 °C) on the warm side of its optimum. The single panel is not lettered.

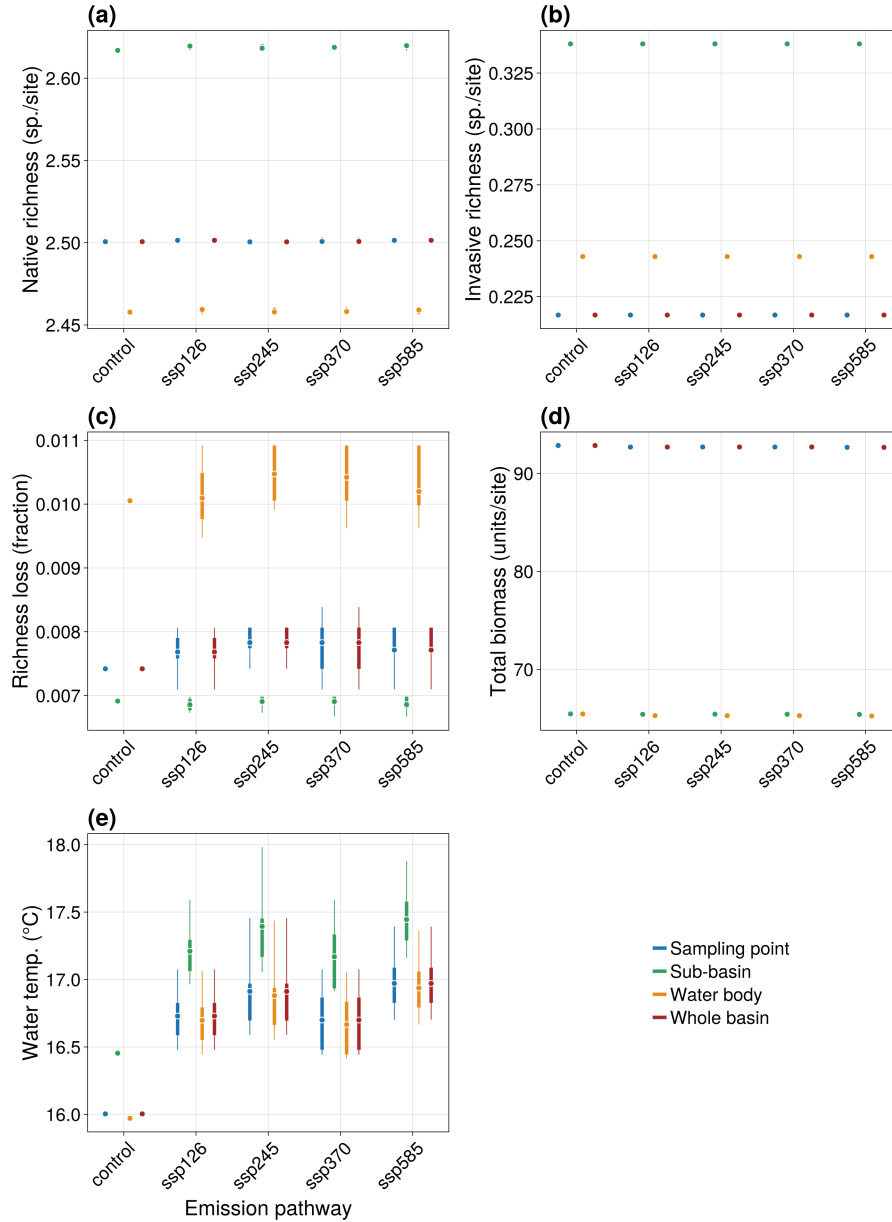


Figure 2: **Final-year (2045) climate-ensemble summary by metric and spatial reporting level.** *Panels:* (a)–(e) are one metric each, labelled on the y-axis with its units: (a) native richness (species per site); (b) invasive richness (species per site); (c) realised native richness loss (fraction of the spun-up baseline); (d) total biomass (model density units per site); (e) projected water temperature ($^{\circ}\text{C}$). The four spatial levels are no longer repeated as separate panels; they are the *colour* inside each panel: blue = sampling point, green = sub-basin, orange = water body, red = whole basin (legend, bottom right). *x-axis:* emission pathway, including the no-warming control. *Points:* mean across the 11 GCMs of that level’s mean across its units. *Bars:* thin = 10–90% and thick = 25–75% spread across GCMs. Merging the four levels into one panel per metric gives them a common y-axis, which is what makes the level-to-level and pathway-to-pathway comparisons direct. *Two nearly flat panels are expected rather than a plotting error:* richness is a threshold count (a species is counted when its density exceeds 0.1). Invasive richness is exactly 0.2168 species per site (168 of the 775×10 invasive site–species combinations) under every scenario and under the control, because no invasive population crosses the 0.1 threshold during 2026–2045; invasive *biomass* does change. Across the 44 climate runs only temperature, and weakly total biomass, separate the pathways; aggregate richness and biomass overlap strongly and the control lies within the scenario spread.

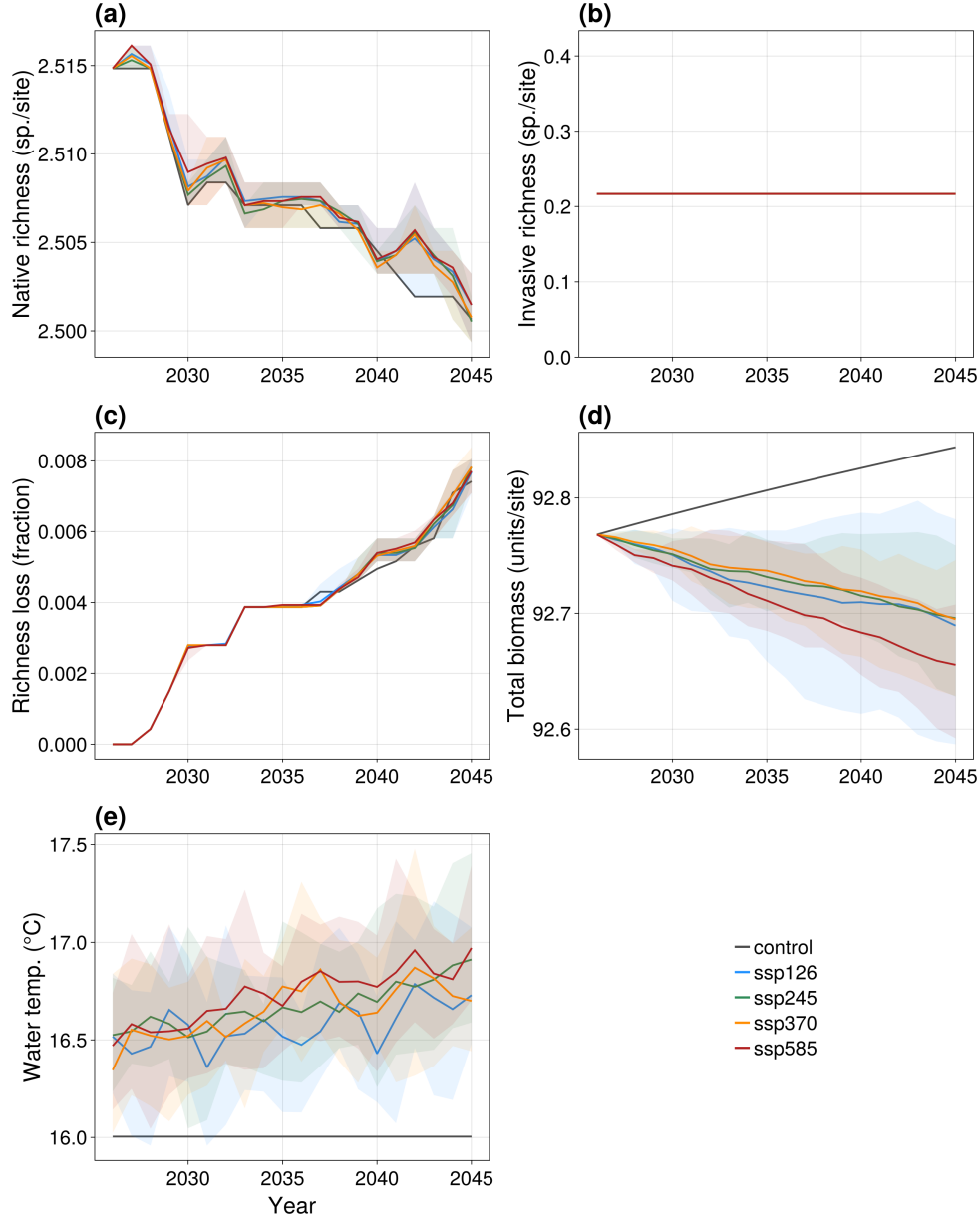


Figure 3: **Annual whole-basin trajectories across emission pathways, 2026–2045.** *Panels:* (a) native richness (species per site); (b) invasive richness (species per site); (c) realised native richness loss (fraction); (d) total biomass (model density units per site); (e) projected water temperature ($^{\circ}\text{C}$). *Lines:* mean across the 11 GCMs of the whole-basin value for each pathway. *Bands:* 10–90% spread across GCMs. *Colour and legend* (outside the panels, bottom right): blue = SSP1-2.6, green = SSP2-4.5, orange = SSP3-7.0, red = SSP5-8.5, black = no-warming control. *How this differs from the forcing-versus-response figure below:* this figure is purely temporal and uses *absolute* water temperature; the forcing-versus-response figure isolates the *warming anomaly* and pairs each trajectory with a final-year response relationship. The temperature trajectories diverge as expected, but native richness, realised richness loss and total biomass remain nearly coincident across pathways, and the control is nearly stationary (its small residual drift is an internal cycle below about 1%). The invasive-richness panel is a flat line at 0.2168 for the threshold-count reason given in the final-year summary figure. The temporal persistence of this pattern supports interpreting the flat aggregate climate slopes as a model result rather than incomplete equilibration.

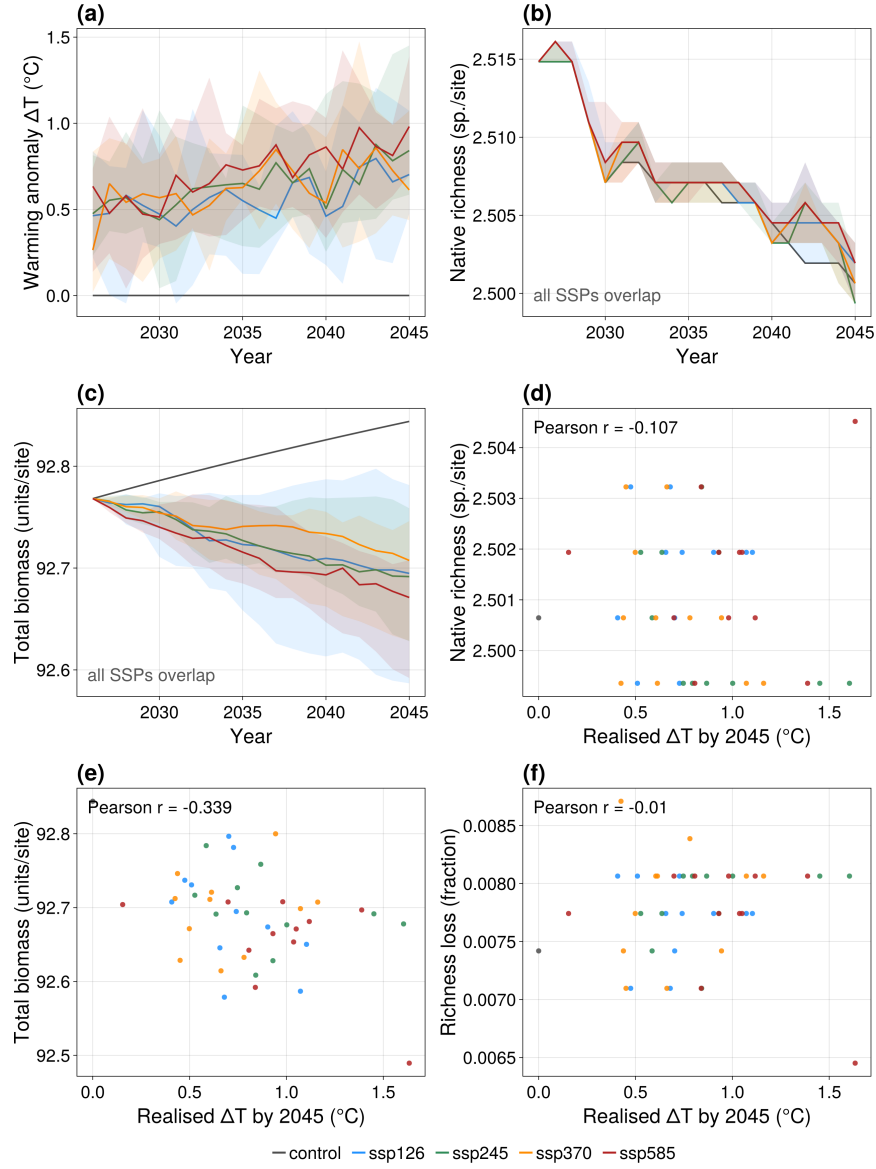


Figure 4: **Forcing versus response.** “Forcing” is the imposed basin-mean water-temperature anomaly ΔT relative to the 1986–2005 baseline, i.e. the warming the climate ensemble adds on top of the observed seasonal climatology; it is shown explicitly because the other figures plot either absolute temperature (the whole-basin trajectory figure) or only the response. *Top row (trajectories, median across the 11 GCMs; band = 10–90% across GCMs):* (a) forcing ΔT ($^{\circ}\text{C}$) over time, with the four SSPs separating; (b) native richness (species per site); (c) total biomass (model density units per site). The annotations “all SSPs overlap” in (b) and (c) flag that the response trajectories lie on top of one another even though the forcing separates. *Bottom row (one point per GCM $\times$ SSP run, coloured by pathway):* (d) final-year native richness, (e) final-year total biomass and (f) final-year relative native richness loss, each against the *realised* warming of that run. The realised anomaly is used here because it is the quantity that actually drives the biology in each run; it differs by GCM from the imposed basin-mean axis used for the ordinary-least-squares slopes in the dose–response figure below. The Pearson correlations quoted in (d)–(f) are computed across the 44 SSP runs; the no-warming control is plotted for reference but excluded from the correlation, matching the dose–response analysis elsewhere in the report. The loss metric is a thresholded realised loss, not an extinction probability. The point of the figure is that the model transmits forcing clearly while the aggregate biological response remains weak.

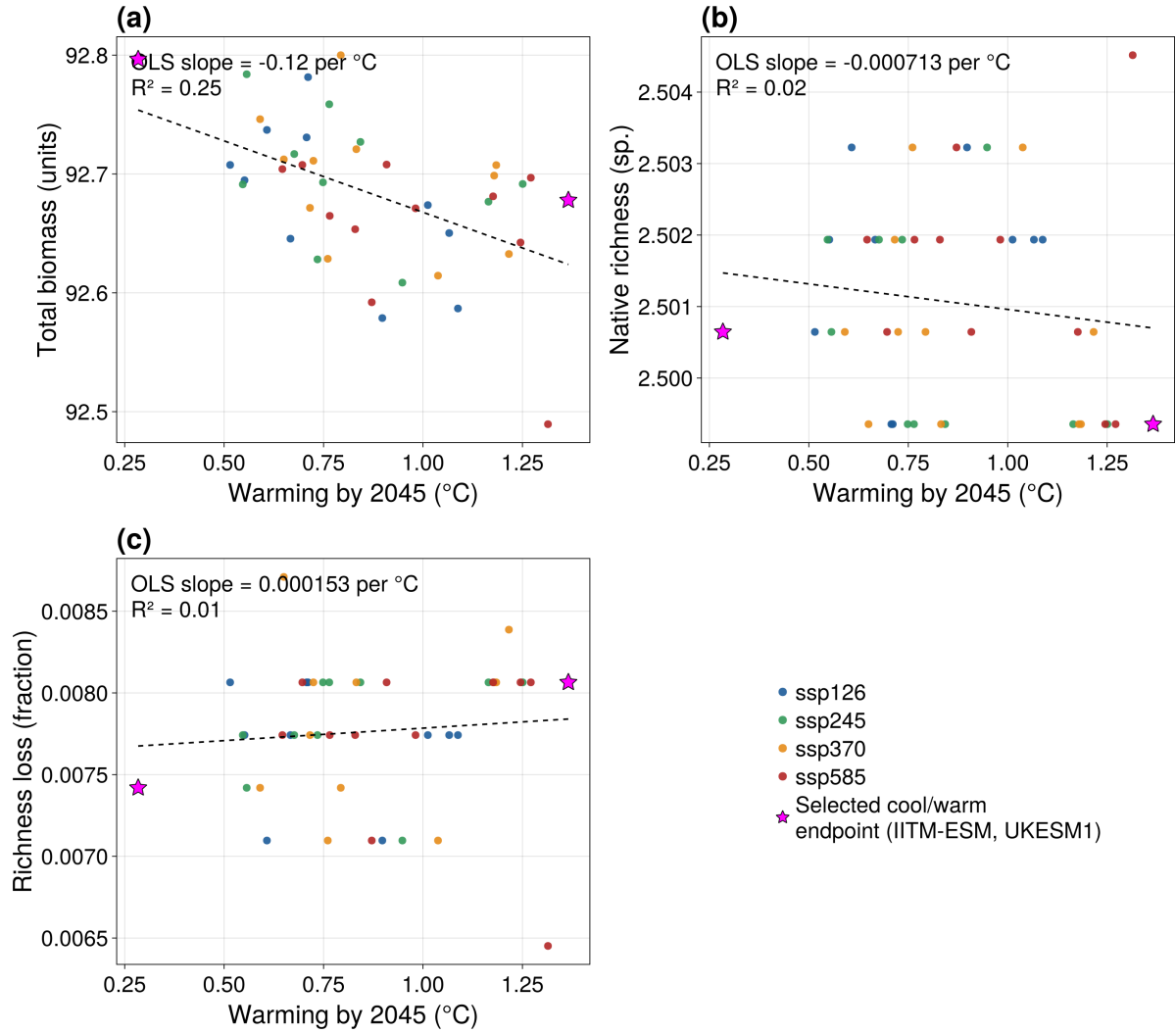


Figure 5: **Basin response versus warming across the 44-run climate ensemble (2026–2045, not end-of-century).** *Panels:* (a) final basin total biomass; (b) final basin native richness; (c) realised native richness loss. *Why there are four points per pathway:* each point is one of the 11 GCMs under one of the four SSPs, giving $11 \times 4 = 44$ points in total; the colour identifies the SSP (legend below the panels). *x-axis:* the imposed basin-mean warming at 2045 (relative to 1986–2005), spanning about 0.28 – 1.37 $^{\circ}\text{C}$. *Quantities shown:* (a) model density units per site, (b) species per site, (c) fraction of spun-up richness. *Dashed black line:* ordinary least-squares fit across the 44 points; the inset gives its slope and R^2 . The fitted slopes are -0.120 model units per $^{\circ}\text{C}$ ($R^2 = 0.25$), -0.00071 species per site per $^{\circ}\text{C}$ ($R^2 = 0.02$) and $+0.00015$ loss units per $^{\circ}\text{C}$ ($R^2 = 0.01$). *Magenta stars:* the two runs selected as the cool endpoint (SSP1-2.6/IITM-ESM) and warm endpoint (SSP2-4.5/UKESM1) of the obstacle experiment. The heavy overlap of pathway colours explains why no emission-pathway ranking is defensible over this horizon.

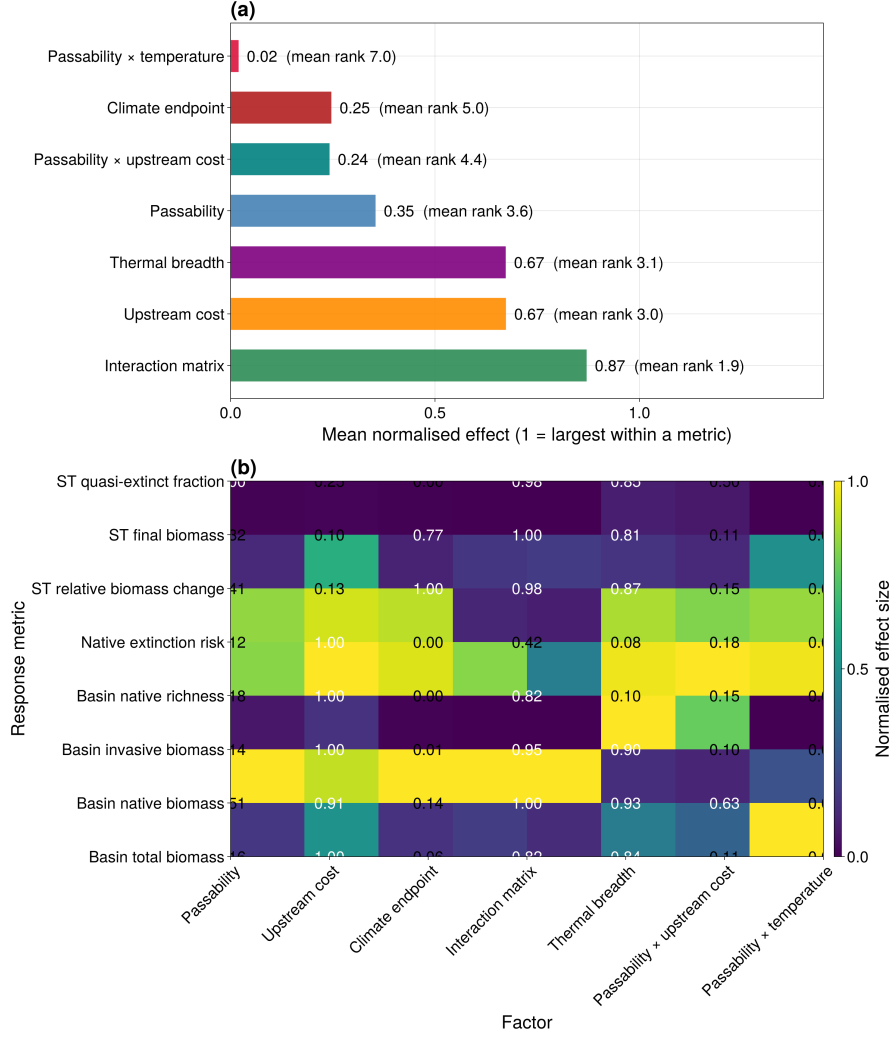


Figure 6: **Factor importance and how it is computed.** *Panel (a)*: overall ranking of the seven factors. The x-axis is the mean *normalised* effect (1.0 = the largest effect for that metric); the annotation gives the mean rank and the mean normalised effect. *Panel (b)*: the per-metric normalised effect size as a heatmap; rows are the eight response metrics, columns the seven factors, the number in each cell is the normalised effect and the colour scale is shown by the colourbar. *Computation*: within each matched factorial block (all other design columns held fixed) the effect span is the difference between the largest and smallest metric value across the levels of one factor; spans are standardised by the standard deviation of that metric across all runs in the sweep, the median span across blocks is taken, and for each metric the medians are divided by the largest so that the most important factor equals 1.0; the rank of a factor is its position within each metric (1 = most important), averaged across the eight metrics. *Factor meanings*: passability = the four obstacle-overlay states; upstream cost = the four elevation-cost settings; climate endpoint = the cool versus warm sensitivity end (only two levels, an external reference axis rather than a full climate-model sample); interaction matrix = original, random placebo and invasive-favouring; thermal breadth = the σ multiplier, confounded with the alternative sweep and its shorter burn-in; the last two columns are the passability × upstream-cost and passability × temperature interactions. Interaction structure ranks first (mean rank 1.88; mean normalised effect 0.871), followed by upstream cost, thermal breadth, passability, the passability–cost interaction, the climate endpoint and the passability–temperature interaction. The apparent importance of passability for brown-trout quasi-extinction is inflated by the near-saturated threshold diagnostic.

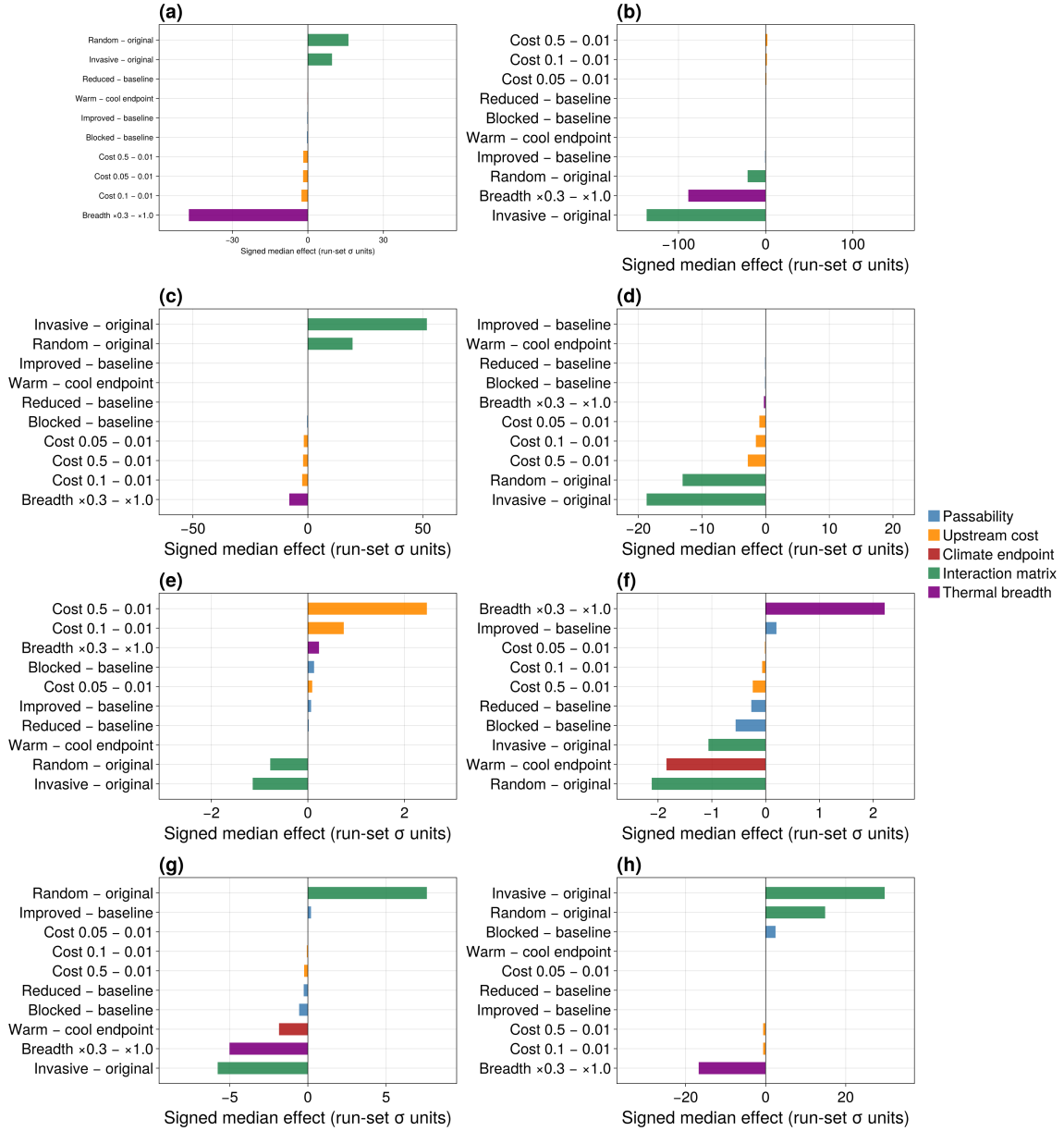


Figure 7: **Signed factor effects (tornado), one panel per response metric.** *Panels:* (a) total biomass, (b) native biomass, (c) invasive biomass, (d) native richness, (e) realised native richness loss, (f) brown-trout relative biomass change, (g) brown-trout final biomass and (h) brown-trout quasi-extinct fraction. *Each bar:* the median over matched factorial blocks of the signed difference between one factor level and its reference level, divided by the standard deviation of that metric across the relevant run set, so all bars are in comparable “ σ units”; a bar to the left means the level lowers the metric. *Bar labels:* every label names the level contrast explicitly, for example “Blocked – baseline” (passability), “Cost 0.50 – 0.01” (upstream movement cost), “Invasive – original” (interaction matrix), “Warm – cool endpoint” (climate) and “Breadth $\times 0.3 - \times 1.0$ ” (thermal breadth); the factor that each contrast belongs to is given by the colour (legend at right). Reference levels are upstream cost 0.01, baseline passability, the original interaction matrix, the cool climate endpoint and breadth multiplier 1.0. *Colour/legend (right):* blue = passability states; orange = upstream cost; red = climate endpoint; green = interaction matrix; purple = thermal breadth. *Reading:* increasing upstream cost generally reduces biomass and richness; passability effects change sign with the background cost; the invasive-favouring matrix has a much larger and more asymmetric effect than the random placebo; the downward thermal-breadth effect is shown for completeness but is confounded with the alternative sweep and its shorter burn-in.

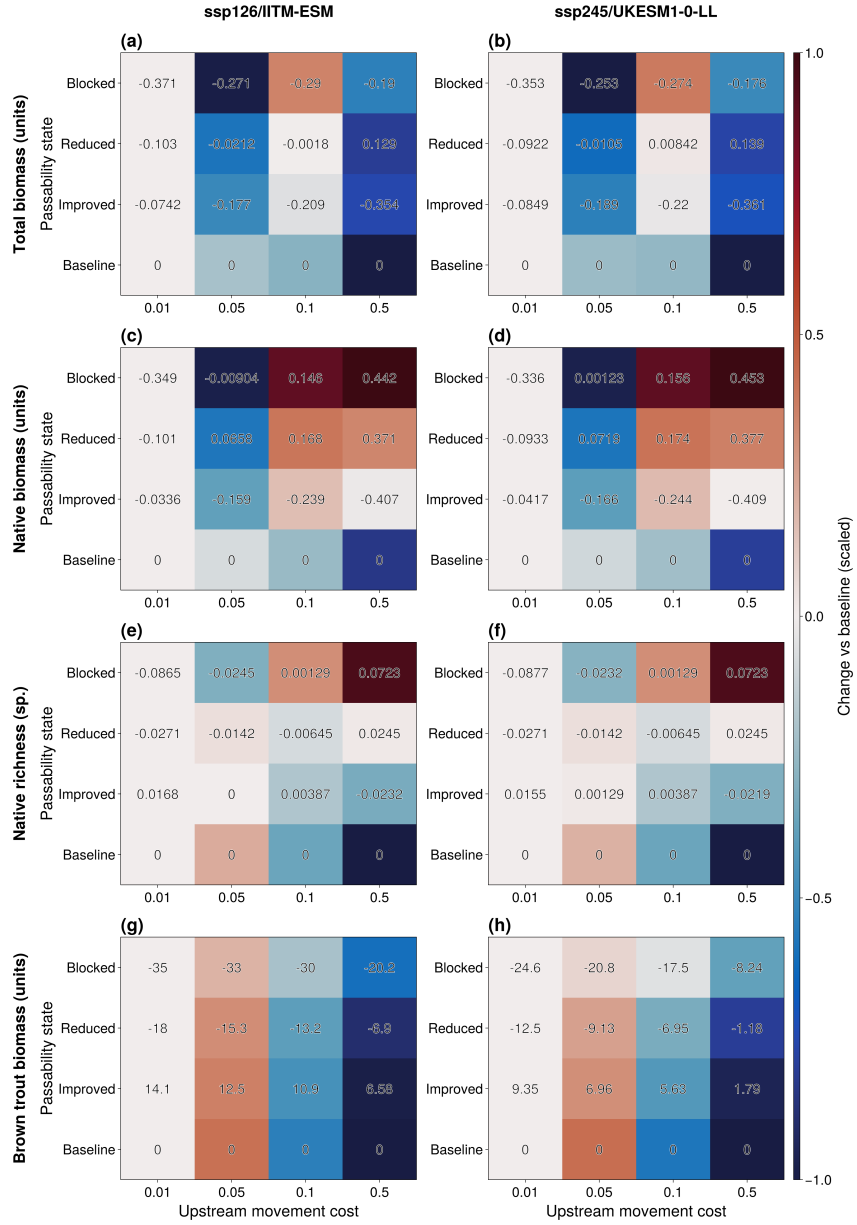


Figure 8: **Passability by upstream-cost interaction (obstacle sweep).** *Panels:* (a)–(h); columns are the two climate endpoints (left: cool, SSP1-2.6/IITM-ESM; right: warm, SSP2-4.5/UKESM1) and rows are four response metrics, labelled on the left: total biomass, native biomass and native richness (all in model density units per site except richness, which is species per site), and brown-trout (ST) final biomass (model units). *Axes:* x = the four upstream movement-cost settings (0.01, 0.05, 0.10, 0.50); y = the four obstacle-overlay passability states (baseline, improved, reduced, blocked). *Cell value:* the raw change of the metric relative to the same-model, same-cost baseline-passability run (so the baseline row is exactly 0 everywhere); units are the model units of the metric. *Cell colour:* the same change divided by that panel's 95th percentile of absolute changes, so colours are comparable within a panel but not between panels (the colourbar spans -1 to $+1$; white = no change). *Reading:* for native biomass the blocked-minus-baseline change goes from -0.349 at cost 0.01 to $+0.442$ at cost 0.50 at the cool endpoint, and the warm endpoint shows the same reversal (-0.336 to $+0.453$); reduced passability follows the same pattern, whereas improved passability is negative at every cost. The sign reversal is context dependence; it does not identify barrier construction as a conservation action. The design cannot by itself separate reduced native rescue from suppressed invasive dispersal.

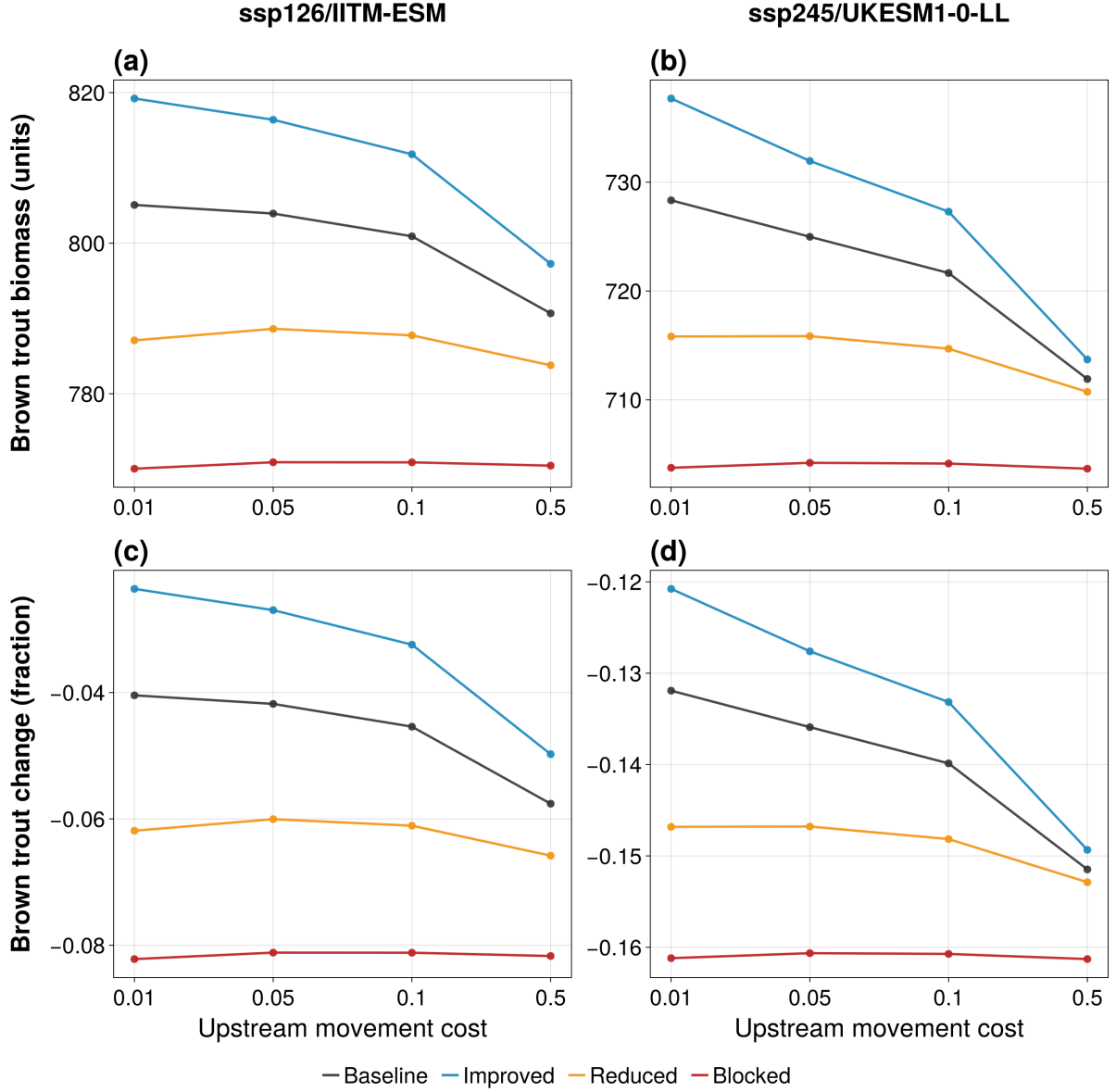


Figure 9: **Brown trout response to hydraulic structure (four equal-size panels).** *Panels:* (a) and (c) are the cool endpoint (SSP1-2.6/IITM-ESM); (b) and (d) are the warm endpoint (SSP2-4.5/UKESM1). The top row is final *Salmo trutta* (ST, brown trout) biomass (model units) and the bottom row is ST relative biomass change (fraction of the spun-up baseline). *x-axis:* the four categorical upstream movement costs (0.01, 0.05, 0.10, 0.50). *Coloured lines/points:* the four passability states; the legend (below the panels) is black = baseline ($\times 1.0$), blue = improved ($\times 1.5$), orange = reduced ($\times 0.5$), red = blocked ($\times 0.1$). *Reading:* all passability curves lie below the spun-up state and the warm endpoint is uniformly more negative; at cost 0.05 the baseline-to-blocked change is 803.9 to 770.9 units at the cool endpoint and 725.0 to 704.2 units at the warm endpoint, so a climate penalty is superimposed on a smaller but non-negligible hydraulic penalty.

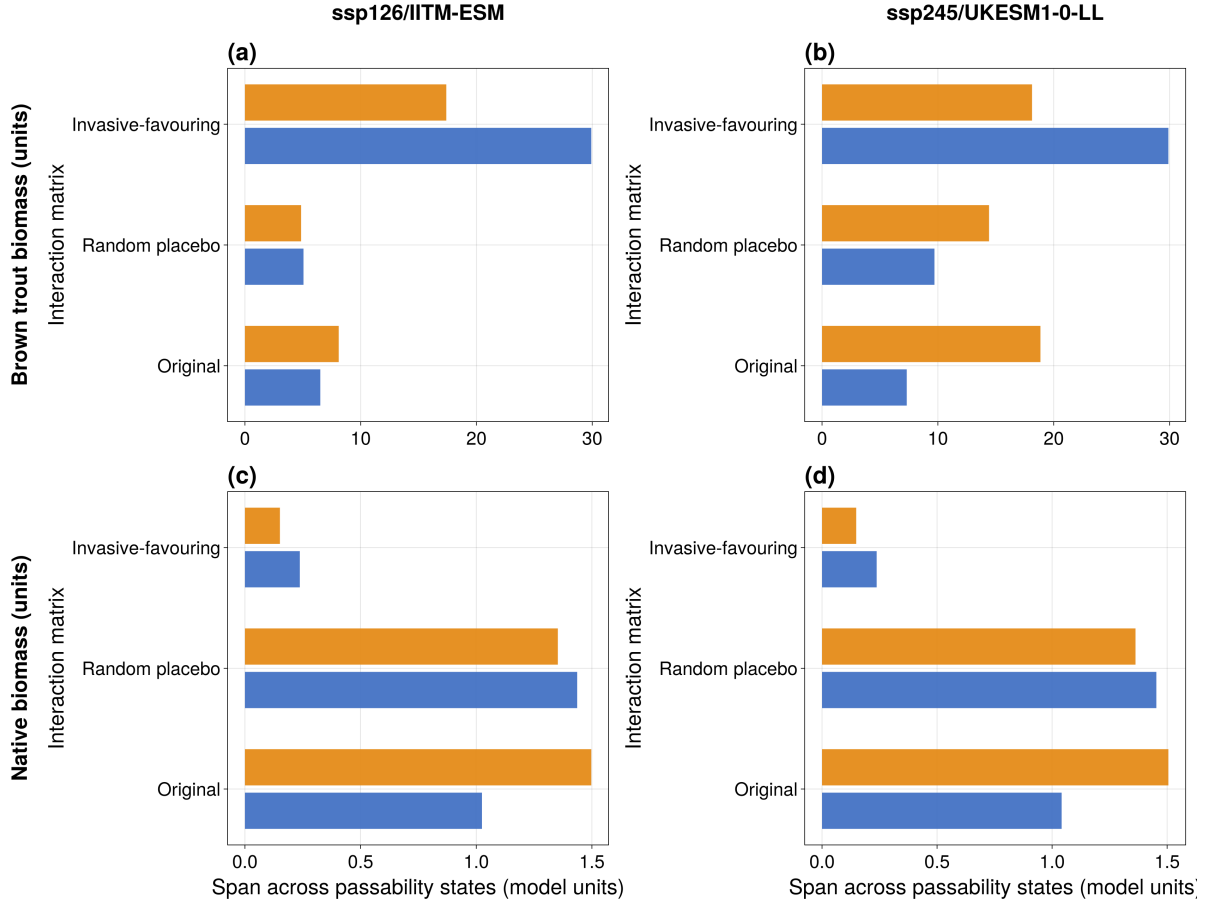


Figure 10: **Passability robustness across interaction matrices (four panels).** *Panels:* (a) and (c) are the cool endpoint; (b) and (d) the warm endpoint (column headers). The top row is brown trout (*Salmo trutta*, ST) final biomass and the bottom row is basin native biomass (all ten native species pooled), both in model units, labelled on the left. *y-axis:* the three interaction matrices (original, random placebo, invasive-favouring). *What each bar is:* the *span across passability* is the difference between the largest and smallest value of the metric when the four passability states (baseline, improved, reduced, blocked) are compared at the same matrix, climate endpoint and cost; a long bar means passability strongly controls the metric. *Why there are two bars per matrix:* each interaction matrix was run at two upstream-cost settings, low (0.01) and high (0.50), drawn as the two bars of each pair; blue bars are the low-cost setting and orange bars the high-cost setting. The panel deliberately carries no legend. *Reading:* ST spans are largest under the invasive-favouring matrix (mean about 23.8 units, versus 10.2 under the original matrix and 8.5 under the random placebo), whereas native-biomass spans are smallest under the invasive-favouring matrix (about 0.2 versus 1.3 and 1.4 units) because native biomass has already collapsed. The figure is a structural-uncertainty warning: both the magnitude and, in this confounded sweep, the sign of passability effects depend on the interaction assumptions.

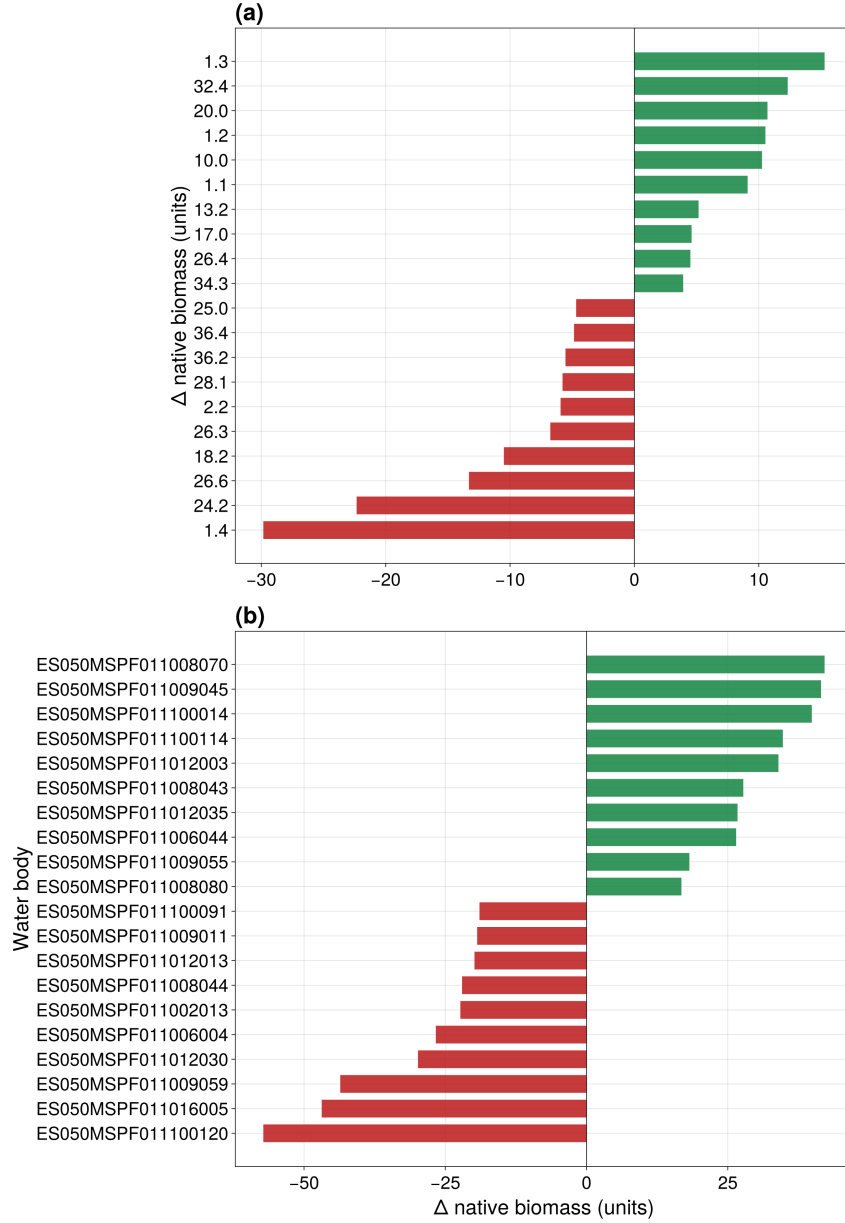


Figure 11: **Spatial distribution of blockage effects.** *Scenario:* warm endpoint, high upstream cost (0.50), year 2045; blocked minus baseline passability. *Panel (a):* the 10 most negative and 10 most positive of the 77 sub-catchments (20 bars); y labels are sub-catchment codes. *Panel (b):* the 10 most negative and 10 most positive of the 289 water bodies (20 bars); y labels are water-body codes, truncated for display. Both panels are the same size, use the same x-axis (change in native biomass, model units) and show bars sorted from most negative to most positive; red bars are losses and green bars are gains. The two panels show only the extremes for legibility; the full distributions are summarised in the text. *Message:* the positive basin-level blocked-minus-baseline effect is a net of local gains and losses, so it must not be read as a uniform benefit of blocking. The sub-catchment or water body is the appropriate scale for management decisions.

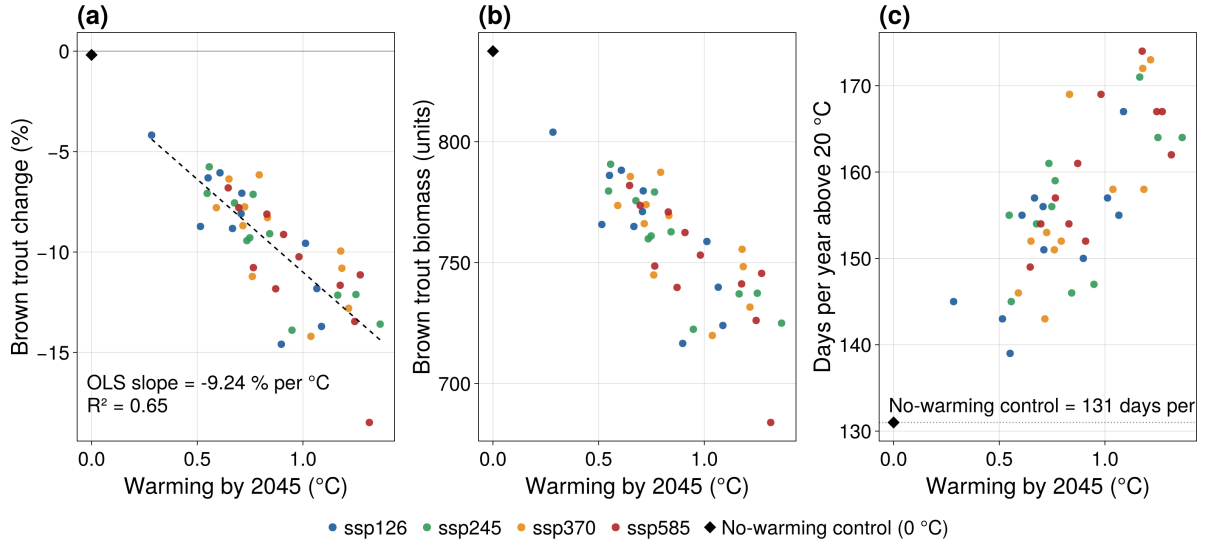


Figure 12: **Brown trout climate dose-response.** *Panels:* (a) relative change in basin *Salmo trutta* (ST) biomass (% of the spun-up baseline); (b) absolute final ST biomass (model units); (c) maximum number of days per year above the empirical 20 °C upper thermal limit. *Points:* one per GCM $\times$ SSP run (44 in total), coloured by pathway; the black diamond at 0 °C is the no-warming control. *x-axis:* imposed basin-mean warming at 2045. *Dashed line in (a):* ordinary least-squares fit, giving a decline of about 9.2% of baseline per °C ($R^2 = 0.65$); the control changes by only -0.2% . (b) shows the absolute decline behind that ratio, and (c) shows the physiological exposure mechanism: maximum annual days above 20 °C rise from 131 in the control (dotted reference line) to 139–174 across scenarios. The figure links a species-level abundance response to a thermal-exposure mechanism while showing why it should not be extrapolated to the whole assemblage.

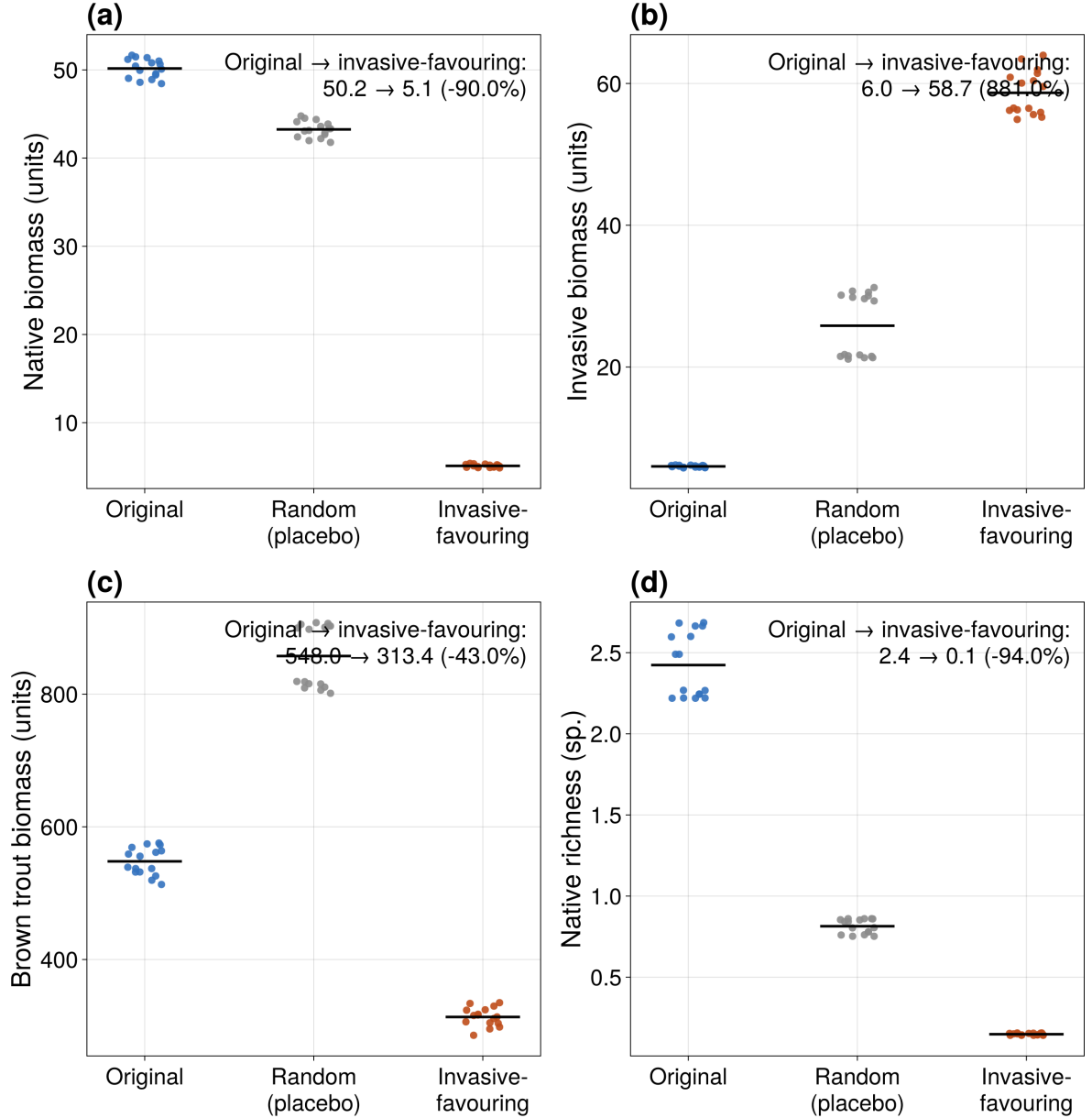


Figure 13: **Interaction-matrix main effects (four panels).** *Panels:* (a) basin native biomass (all native species pooled); (b) basin invasive biomass; (c) brown trout (*Salmo trutta*, ST) final biomass; (d) basin native richness (species per site). All biomass panels are in model units; the y-axis label gives the units. *x-axis:* the three interaction assumptions, original qualitative matrix, random placebo matrix and invasive-favouring matrix. *Points:* every one of the 16 runs per matrix (two climate endpoints $\times$ two upstream costs $\times$ four passability states); a small horizontal jitter separates overlapping points. *Black horizontal line:* the matrix mean across those 16 runs. *Annotations:* the mean original $\rightarrow$ invasive-favouring change with its percentage. Relative to the original matrix, the invasive-favouring matrix shifts native biomass from about 50.2 to 5.1 units (-90%), ST biomass from 548 to 313 (-43%), native richness from 2.4 to 0.1 (-94%) and invasive biomass from 6.0 to 58.7 ($+881\%$), while total biomass changes comparatively little (56.2 to 64.2; not shown here). The random-matrix means are not ecologically interpretable because some species have near-zero burn-in baselines. The figure therefore shows redistribution between community compartments, but the deliberately extreme matrix, the confounded thermal-breadth multiplier and the shorter burn-in make it a bounding scenario rather than a forecast.

11 Summary Tables

Table 2: Climate-ensemble response at basin scale.

Quantity	No-warming control	Climate-ensemble response
Added basin-mean warming by 2045	0.00 °C	0.74 (SSP1-2.6), 0.87 (SSP2-4.5), 0.88 (SSP3-7.0), 0.97 (SSP5-8.5) °C
Total biomass change	+0.08%	−0.120 model units per °C; $R^2 = 0.25$
Native biomass change	+0.10%	Weak negative response; no defensible pathway ranking
Native richness change	−0.57%	−0.00071 species per site per °C; $R^2 = 0.02$
Native richness-loss indicator	< 1%	+0.00015 units per °C; $R^2 = 0.01$; never above approximately 0.9%
Trout biomass change	−0.2%	−9.0% (SSP1-2.6) to −10.9% (SSP5-8.5)

Table 3: Principal within-experiment variance contributions.

Response	Dominant factor	SS	Other notable contributions
Total biomass	Upstream cost	0.961	Climate endpoint: 0.006
Native biomass	Upstream cost	0.578	Passability $\times$ cost: 0.242; passability: 0.155; climate endpoint: 0.026
Native richness	Upstream cost	0.947	Climate endpoint: approximately zero
Invasive biomass	Upstream cost	0.970	Interaction structure is also large in the alternative sweep
Trout final biomass	Climate endpoint	0.886	Passability: 0.090; upstream cost: 0.011

12 Data Sources

This final report is a synthesis of the current deterministic model outputs and empirical input collections. The principal data sources are fish-density and juvenile observations, site connectivity and environmental measurements, species thermal-trait data, qualitative interaction information, the 2026 obstacle inventory and daily downscaled water-temperature projections for the Guadalquivir basin. Persistent data/code accessions, licences and any restrictions on sensitive spatial data should be added in the submission version.

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